

Summary of Findings - Alt Mobility Data Analyst Assignment

Overview

As part of this internship assignment for Alt Mobility, I explored two datasets - one containing customer orders and another containing payment records. The goal was to use SQL and Power BI to find useful insights about sales trends, customer behavior, and payment performance, and to visualize customer retention.

What I Found (Task-wise Insights)

Task 1: Order and Sales Analysis

Most orders had a status like 'delivered', 'shipped', or 'pending'. I calculated the total revenue by summing all order amounts. I also tracked monthly revenue, and noticed that some months had more sales than others, which might suggest seasonal trends.

Task 2: Customer Analysis

Many customers ordered only once, but there's a fair number of repeat customers too. I found which customers ordered more than once, and also counted new customers each month. This helped understand customer loyalty and growth.

Task 3: Payment Status Analysis

Most payments were successful, but there were still several failed transactions. I looked at how these failed payments changed over time - which could help improve the payment system. This might be due to problems with certain payment methods like PayPal or credit cards.

Task 4: Order Details Report

I combined both datasets using a LEFT JOIN in SQL to create a full report. This report shows each order with its payment status (if available), which helps identify unpaid or failed orders. It would be useful for customer support or the finance team.

Task 5: Customer Retention Visualization

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I used Power BI to build a heatmap showing how many customers came back to order in later months. This is based on the month they first ordered (called the cohort month). I noticed that most customers don't return after the first month, but some cohorts had better retention than others.

What I Recommend

- Try to bring back one-time buyers by offering discounts or loyalty points.
- Look into why payments are failing - maybe improve the payment process or send payment reminders.
- Study the best-performing customer groups to improve marketing strategies.
- Use dashboards to track important things like revenue, repeat customers, and payment success more easily.

Tools I Used

- SQL: MySQL Workbench to write and test queries
- Power BI: For creating the cohort retention heatmap
- GitHub: To store and organize all my code and results